

termesures such as flares. Extreme close-in agility of IRIS-T with capability to make 60g turns at 60°/s allows the missile to engage targets even behind the launching aircraft. It was inducted into service in 2005.

Vympel R-73 (NATO designation AA-11 Archer), manufactured by Tbilisi Aircraft Manufacturing, is also a short-range air-to-air missile with maximum speed of 2.5Mach and maximum operational range of 20km (R-73E), 30km (R-73M1) and 40km (R-73M2). It employs a cryogenically-cooled all-aspect IR homing seeker with high off-bore sight capability, allowing the missile to see 40° off the missile's centre line. It was inducted into service in 1982. R-73 is also on the inventory of Indian Air Force.

MICA-IR (Fig. 18), manufactured by MBDA, is a short- and medium-range air-to-air missile having a maximum operational range of 50km and a maximum speed of 3Mach. It uses an IIR seeker that gives the missile high resistance to countermeasures such as chaff and decoy flares. It can lock on after launch, which means that it can engage targets outside the missile's acquisition range at the time of launch. It is in service since 2000. Indian Air Force has ordered Mica-IR missiles for its MIRAGE upgrade 2000H multi-role fighters.

AIM-132 ASRAAM is a short-range air-to-air missile manufactured by MBDA. It uses an IIR seeker with lock-on after launch capability, has a maximum speed of 3+ Mach and maximum operational range of 50km. It is in use in Royal Air Force and Royal Australian Air Force having replaced AIM-9 Sidewinder. Indian Air Force is also acquiring ASRAAM to replace the ageing Matra Magic missiles. These missiles will be integrated on Jaguar strike aircraft.

AIM-9X (Fig. 19) is the latest addition to the Sidewinder family of short-range air-to-air missiles developed by Raytheon Co. It features an IIR seeker focal plane array seeker with off-bore sight capability of 90°. The IIR seeker gives it higher



Fig. 18: MICA-IR air-to-air missile



Fig. 19: AIM-9X air-to-air missile

resistance to IR countermeasures. The first Sidewinder missile was developed in the 1950s. AIM-9X is the fifth-generation Sidewinder and is now in production. AIM-9X uses passive IR energy emitted by target aircraft for acquisition and tracking, which provides a launch-and-leave air combat missile capability. AIM-9X Sidewinder is characterised by an operational range of about 35km and a speed of 2.5Mach.

AIM-9X Block-I was the first in the family of these missiles. Currently, AIM-9X Block-II has entered full-scale production. Block-II missiles are the upgraded version of Block-I missiles with lock-on-after-launch being the main added feature. The development work has commenced on AIM-9X Block-III missiles. Block-III missiles will have 60 per cent longer range and use insensitive munitions warhead for increased ground crew safety in addition to replacing old components with state-of-the-art ones. Block-III Sidewinder missiles are expected to achieve operational capability by 2022.

Python-5 is one of the most advanced air-to-air missiles in the world. Different variants of Python family include Shafrir-1, Shafrir-2, Python-3, Python-4 and Python-5.

Python-5 is the latest addition to the family and is the fifth-generation

air-to-air missile. Manufactured by Rafael Advanced Defence Systems in Israel, it has many advanced features such as an IIR seeker to give it high immunity to IR countermeasure, target lock-on before and after launch capability to engage targets beyond visual range, higher kill probability and revolutionary full-sphere-launch envelope from very short to beyond visual ranges. It can lock on to the target after launch even when the target is 100° off the bore sight. The missile has an operational range of more than 20km and a speed of 4Mach.

Modern IR-guided missiles using IIR seekers with advanced digital signal processing techniques have much wider detection angles, giving them the capability to launch missiles from large off-bore sight angles. Helmet-mounted sights with pilots of the launch aircraft allow them to distinguish between the target aircraft and a point source of intense heat, such as a flare. These missiles almost invariably have lock-on-after-launch feature, enabling them to engage targets from a very small range to beyond the visual range.

Another recent advancement in missile guidance is the use of electro-optical imaging. The electro-optical seeker scans the designated area for targets via optical imaging. Once the target is acquired, the missile locks-on to it for the kill. Electro-optical seekers can be programmed to hit the designated spot on the target aircraft. The designated spot could be the most vulnerable point of the target. Since electro-optical imaging does not depend on the target aircraft's heat signature, it can be used against low-heat targets such as unmanned aerial vehicles and cruise missiles.

There have been advances in control systems for better manoeuvrability of the flight path. Crew safety on ground has been another concern and it has led to the development of insensitive munitions warheads that do not detonate accidentally.

*To be concluded next month*